qudossl crypto benchmark

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qudossl (OpenSSL 3.5.7 with ML-KEM and ML-DSA delegated to qudo-pqc) against pristine OpenSSL 3.5.7, measured with the benchmark tools published by OpenSSL and by the rustls project, each built once against each tree.

Instruction counts

`instructions.csv` absent - run `./scripts/run-instr.sh`.

Primitives

`openssl speed`, the tool OpenSSL ship for this purpose. These cells isolate the delegated math: no classical algorithm is involved, so the whole measured difference is ML-KEM or ML-DSA.

mode	algorithm	operation	qudossl op/s	openssl op/s	ratio
default	ML-KEM-1024	keygen	86,466	40,363	2.142x
default	ML-KEM-1024	encaps	110,321	78,106	1.413x
default	ML-KEM-1024	decaps	84,004	52,824	1.591x
default	ML-KEM-512	keygen	170,814	96,084	1.778x
default	ML-KEM-512	encaps	244,262	159,685	1.530x
default	ML-KEM-512	decaps	186,739	104,414	1.788x
default	ML-KEM-768	keygen	115,392	61,131	1.888x
default	ML-KEM-768	encaps	155,872	110,265	1.414x
default	ML-KEM-768	decaps	121,236	71,580	1.695x
default	SecP256r1MLKEM768	keygen	55,570	39,097	1.421x
default	SecP256r1MLKEM768	encaps	22,424	21,196	1.058x
default	SecP256r1MLKEM768	decaps	25,522	22,285	1.145x
default	SecP384r1MLKEM1024	keygen	2,639	2,521	1.047x
default	SecP384r1MLKEM1024	encaps	1,387	1,348	1.029x
default	SecP384r1MLKEM1024	decaps	2,707	2,593	1.044x
default	X25519MLKEM768	keygen	42,858	32,153	1.333x
default	X25519MLKEM768	encaps	27,253	25,543	1.067x
default	X25519MLKEM768	decaps	39,174	32,559	1.204x
default	ML-DSA-44	keygen	34,280	23,190	1.478x
default	ML-DSA-44	sign	10,940	4,151	2.636x
default	ML-DSA-44	verify	38,768	21,428	1.809x

mode	algorithm	operation	qudossl op/s	openssl op/s	ratio
default	ML-DSA-65	keygen	16,936	12,269	1.380x
default	ML-DSA-65	sign	7,030	2,566	2.740x
default	ML-DSA-65	verify	23,826	13,980	1.704x
default	ML-DSA-87	keygen	12,921	9,140	1.414x
default	ML-DSA-87	sign	5,648	2,157	2.619x
default	ML-DSA-87	verify	14,503	8,804	1.647x
fips	ML-KEM-1024	keygen	30,063	17,515	1.716x
fips	ML-KEM-1024	encaps	109,852	78,732	1.395x
fips	ML-KEM-1024	decaps	83,835	53,509	1.567x
fips	ML-KEM-512	keygen	63,418	37,797	1.678x
fips	ML-KEM-512	encaps	243,426	162,167	1.501x
fips	ML-KEM-512	decaps	186,141	104,625	1.779x
fips	ML-KEM-768	keygen	42,045	25,376	1.657x
fips	ML-KEM-768	encaps	155,459	110,507	1.407x
fips	ML-KEM-768	decaps	120,056	72,945	1.646x
fips	SecP256r1MLKEM768	keygen	12,014	10,024	1.199x
fips	SecP256r1MLKEM768	encaps	10,453	10,181	1.027x
fips	SecP256r1MLKEM768	decaps	25,729	22,433	1.148x
fips	SecP384r1MLKEM1024	keygen	564	545	1.037x
fips	SecP384r1MLKEM1024	encaps	475	473	1.004x
fips	SecP384r1MLKEM1024	decaps	2,684	2,623	1.023x

mode	algorithm	operation	qudossl op/s	openssl op/s	ratio
fips	X25519MLKEM768	keygen	27,033	18,874	1.432x
fips	X25519MLKEM768	encaps	27,672	25,888	1.069x
fips	X25519MLKEM768	decaps	40,619	33,182	1.224x
fips	ML-DSA-44	keygen	6,770	2,984	2.269x
fips	ML-DSA-44	sign	11,046	4,148	2.663x
fips	ML-DSA-44	verify	38,745	21,443	1.807x
fips	ML-DSA-65	keygen	4,111	1,797	2.289x
fips	ML-DSA-65	sign	7,087	2,541	2.790x
fips	ML-DSA-65	verify	23,934	13,907	1.721x
fips	ML-DSA-87	keygen	3,076	1,469	2.094x
fips	ML-DSA-87	sign	5,640	2,168	2.602x
fips	ML-DSA-87	verify	14,546	8,788	1.655x

54 cells measured. 44 showed a consistent difference in every repetition, spanning 1.15x to 2.79x.

TLS handshake

OpenSSL's `perftools/handshake`: a combined in-memory client and server, no socket. Lower microseconds is better, so the ratio is inverted to keep "higher is better for qudossl" consistent with the other tables.

mode	group	qudossl us	openssl us	ratio
default	SecP256r1MLKEM768+ML-DSA-44	384.9	587.9	1.528x
default	X25519+ECDSA	243.7	241.9	0.993x
default	X25519+ML-DSA-44	322.4	498.8	1.547x
default	X25519MLKEM768+ML-DSA-44	358.0	557.4	1.557x
fips	X25519MLKEM768+ML-DSA-44	382.0	588.9	1.542x

Cells are named `<key exchange>+<signature>`.

- **`X25519+ECDSA`** - the negative control. Nothing is delegated on either side, so it must read parity.
- **`X25519MLKEM768+ML-DSA-44`** - the deployment case: both the key exchange and the signature are delegated.
- `X25519+ML-DSA-44` isolates the signature. Note it still shows a large difference despite the classical key exchange, because the ML-DSA certificate is delegated - which is why it is not a control.

Session resumption

`resumption.csv` absent.

Memory

`memory.csv` absent.

Limitations

	measured on
CPU	Apple M4
architecture	arm64
SIMD available	neon
cores	10
OS	macos
kernel	Darwin 25.5.0
measured	2026-08-24T07:08:25Z

The two builds

Both report the same version, because qudossl **is** OpenSSL 3.5.7 with the ML-KEM and ML-DSA cores delegated to qudo-pqc; the version string is deliberately not rebranded. They are separate trees, built and linked independently.

	qudossl	openssl
version	OpenSSL 3.5.7+qudo-1.0.0 9 Jun 2026 (Library: OpenSSL 3.5.7+qudo-1.0.0 9 Jun 2026)	OpenSSL 3.5.7 9 Jun 2026 (Library: OpenSSL 3.5.7 9 Jun 2026)

- **Wall-clock figures are host-dependent.** CPU frequency, core type and thread placement all affect them and none is controlled here. The instruction counts are far more stable - about 2% - though not exact, for the reason given in that section.
- **No network is involved anywhere in this suite.** Every figure is an in-process measurement. Deployment behaviour under real traffic is not characterised here.
- `openssl speed -seconds 2` is used; upstream's default for this operation class is 10 s. More repetitions at 2 s were preferred to fewer at 10 s, because a disturbed window is a disturbed sample at either length.
- **arm64, SIMD available: neon.** qudo-pqc selects its backend at runtime, so only the path this CPU offers was exercised. Other architectures, and any SIMD level absent above, are not covered by these ratios.

Provenance

file	rows	sha256
`environment.csv`	13	`9d8e3abace0575688dd44948739fac72d64db19a613781c7259ecd4206dd3ef6`
`handshake.csv`	60	`4e253ac5d0cc773770bf4b7e9bed494c37be514ea7081506817606cdc30ca66c`
`primitives.csv`	216	`52ef3b82ef1a08ce1d1e6c6aa00eaa55db687e4f6b1884b9730c912726fe8bc2`

Tools: OpenSSL perftools and ctz/openssl-bench, both vendored under `benchmark/`, each built once against each OpenSSL tree.

Generated 2026-08-24T07:38:44Z on Darwin 25.5.0 arm64.